

Ankita Joshi

Postdoctoral Researcher · Cincinnati Children's Hospital Medical Center

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Education

University of Georgia

PhD in Computer Science (GPA: 3.72/4.0)

Athens, Georgia

August 2017 - Dec 2022

- Advisor: Dr. Yi Hong
- Research Fellow (tuition + stipend)
- Dissertation: Developed population-based image metamorphosis and registration models using deep learning that account for brain structure and image appearance changes simultaneously.
- Interests: Medical image analysis, Deep learning, Machine learning, Statistical analysis.

University of Georgia

Masters in Artificial Intelligence (GPA: 3.66/4.0)

Athens, Georgia

January 2015 - Dec 2017

- Advisor: Dr. Shannon Quinn
- Research Fellow (tuition + stipend)
- Thesis: Developed an open-source implementation of a distributed version of the Spectral Clustering algorithm using Apache Spark which works on big data.
- Interests: Image processing, Clustering, Classification, feature engineering/selection, Dimensionality reduction, topic modeling.

University of Pune

Bachelors in Computer Science & Engineering (GPA: 4.0/4.0)

Pune, India

August 2009 - July 2013

- Top 1% of class
- Courses: Algorithms; Data Structures; OS; Data Mining; NLP; DBMS; Big Data

Experience

Cincinnati Children's Hospital Medical Center

Postdoctoral Researcher

Cincinnati, Ohio

January 2023 – Present

PIs: Dr. Lili He, Dr. Nehal Parikh

- Developed statistical methodology and process (hypothesis testing, generalized linear modeling, multiple-testing correction, regression models, correlation analysis, functional data analyses).
- Led development of predictive models linking imaging features with developmental and clinical outcomes.
- Collaborated with cross-disciplinary teams including radiologists, neuroscientists, and AI researchers. Conducted research using high-performance computing (HPC) clusters for large-scale biomedical data processing and model development.

AT&T Labs

Research Intern – Technical II

Remote (Bedminster, New Jersey)

May 2020 – Aug 2020

Mentors: Yuan Ding, Pradeep Bansal

- Developed machine learning models on network traffic (packets and their characteristics) to identify traffic in the network (e.g., movie trailers on YouTube, Netflix traffic) and the quality of experience (QoE) of the traffic and submitted patent for the same.
- Deployed ML models built using the packet flow data without looking at the actual content inside; since it is encrypted. **Patent Pub. No.:** US 2023/0334287 A1

CareerBuilder

Data Science R&D Intern

Atlanta, Georgia

August 2016 – May 2017

Hosts: Dr. Qiaoling Liu

- Collaborated with the R&D team to develop an algorithm for the “Named Entity Recognition” task to help link employer names in job postings to entities in an employer knowledge base. (employee name normalization)
- Contributed to the training and evaluation of the cluster-level normalization algorithm.

Sikka.ai

AI Scientist Intern

San Jose, California

June 2016 – August 2016

Hosts: Yelena Meroyan

- Conceptualized, designed and implemented a proof-of-concept (POC) up to production for developing transactional bots for retail health care.

Publications

Journals

- 2025 FODSeg – Tract-Specific White matter Segmentation Using A Complete Angular Distribution
Ankita Joshi, Hailong Li, Nehal A. Parikh, Lili He
Under Review
- 2025 Mapping white matter microstructure at term age to motor outcomes at 2 years in very preterm infants: a multicenter cohort study.
Ankita Joshi, Wei Jia, Junqi Wang, Hailong Li, Mekibib Altaye, Nehal A. Parikh, Lili He
Under Review
- 2024 A Systematic Review of Automated Methods to Perform White Matter Tract Segmentation [Link]
Ankita Joshi, Hailong Li, Nehal A. Parikh, Lili He
Frontiers in Neuroscience, Brain Imaging Methods
- 2023 R2Net: Efficient and flexible diffeomorphic image registration using Lipschitz continuous residual networks. [Link] [Code]
Ankita Joshi, Yi Hong
Medical Image Analysis

Peer-reviewed Conferences

- 2025 Enhancing zero-shot classification in chest x-rays via cross-modal representation and attention alignment. [Link]
Redha Ali, Almabrok Essa, **Ankita Joshi**, Fatmah Ali, EEIsmoatazbellah Ben Omar, Adel Alshamili, Russell C. Hardie
SPIE Conference, Multimodal Image Exploitation and Learning 2025
- 2023 MetaRegNet: Metamorphic Image Registration Using Flow-Driven Residual Networks. [Link] [Code]
Ankita Joshi, Yi Hong
Computation Mathematics Modeling in Cancer Analysis (CMMCA) at MICCAI
- 2022 Diffeomorphic Image Registration using Lipschitz Continuous Residual Networks. [Link] [Code]
Ankita Joshi, Yi Hong
Medical Imaging with Deep Learning (MIDL)
- 2022 Efficient Diffeomorphic Image Registration Using Multi-Scale Dual-Phased Learning. [Link]
Ankita Joshi, Yi Hong
IEEE 19th International Symposium on Biomedical Imaging (ISBI)
- 2021 Multi-Resolution Diffeomorphic Image Registration Using Efficient Flow Field Estimation. [Link] [Code]
Ankita Joshi, Yi Hong
Med-NeurIPS
- 2017 A Supporting Employer Name Normalization at both Entity and Cluster Level. [Link]
Qiaoling Liu, Faizan Javed, Vachik Dave, **Ankita Joshi**
SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)
- 2016 Particle Swarm Optimization.
Sanjay Sarma, **Ankita Joshi**
Sixteenth Annual Early Career Technical Conference

Technical Skills

Languages	Python, Scala, MATLAB, R Java, C, Prolog
ML/DL Libraries	PyTorch, TensorFlow, Keras, numpy, scipy, scikit-learn, Nvidia cuDNN
Big Data	Apache Hadoop, Apache Spark, Amazon EMR, HPC Cluster
OS	Linux, Unix, Windows
Imaging Platforms	SimpleITK, nibabel, MRtrix3, FSL, ANTs, TrackVis, DTIStudio, 3DSlicer
Version Control	Git

Teaching & Mentorship

Teaching

Fall 2022	Unix Basics, C Programming, Teaching lab, Designed Exams and Assignments, Assignment Grading, Project Mentorship	TA
Spring 2021	Operating Systems (Graduate), Teaching lab, Assignment Grading, Project Mentorship	TA
Spring 2018	Systems Programming (Undergraduate), Teaching lab, Designed Exams and Assignments, Assignment Grading, Project Mentorship	TA
Fall 2017	Systems Programming (Undergraduate), Teaching lab, Designed Exams and Assignments, Assignment Grading, Project Mentorship	TA

Mentorship

Fall 2024	Jinzhao Qian, UC PhD Student
Fall 2022-Summer 2023	Pooja Reddy, UGA MS Student
Fall 2022	Eno Bempah, UGA Undergraduate Student
Fall 2022-Summer 2023	Ruthik Kale, UGA MS Student
Spring 2015-Fall 2016	Sayali Kale, UGA MS Student

Invited Talks

April 2025	Mapping white matter microstructure at term age to motor outcomes at 2 years in very preterm infants, Imaging Radiology Center	CCHMC
Summer 2022	Diffeomorphic Image Registration using Lipschitz Continuous Residual Networks, Medical Imaging with Deep Learning	Remote
Spring 2018	Efficient diffeomorphic image registration using multi-scale dual-phased learning, IEEE International Symposium on Biomedical Imaging	Remote

Service

Fall 2024	Reviewer, IEEE ISBI, ACM ICMI 2025, Medical Image Analysis, Computers in Biology and Medicine
Spring 2018	Organizer, UGA Computer Science Research Fair
Fall 2022	Board Member, UGA CS Graduate Student Association, Indian Student Association

References

Dr. Lili He Associate Professor	<i>Cincinnati Children's Hospital Medical Center</i> lili.he@cchmc.org
Dr. Hailong Li Assistant Professor	<i>Cincinnati Children's Hospital Medical Center</i> hailong.li@cchmc.org
Dr. Nehal Parikh Professor, UC Department of Pediatrics	<i>Cincinnati Children's Hospital Medical Center</i> nehal.parikh@cchmc.org
Dr. Yi Hong Associate Professor	<i>Shanghai Jiao Tong University</i> yi.hong@sjtu.edu.cn
Dr. Suchendra Bhandarkar Professor	<i>University of Georgia</i> suchi@uga.edu